

Predicting Hit Songs Using Repeated Chorus

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Question and dataset

Can a 15-second chorus distinguish a source year-end Billboard hit (class 1) from another chart song (class 0)? Both classes can have charted. The public feature-table task differs from the supplied reference paper. This update preserves its labels and original historical-test membership.

The audited table contains 751 songs, 71 provided artist names and 518 raw audio features. Class counts are 385 other-chart songs and 366 year-end hits. Original recordings are unavailable. Names are grouping metadata, never model inputs.

Controlled evaluation

The original development partition has 597 songs from 52 artist-name groups. We compare 35 predeclared settings with fixed seeds. 5 outer folds each use 3 inner folds for nested selection. The full development partition then selects the final candidate using inner-fold balanced accuracy. Historical labels never tune it.

All learned imputation, filtering, scaling, PCA and supervised selection fit inside training folds. The bounded search retains the original classifier controls, adds Extra Trees, and compares unreduced features, PCA, ANOVA selection, seeded mutual information and a declared feature-family subset. Native estimator thresholds remain fixed.

Audit and implementation

The audit reproduced all original recorded metrics. Repairs bind audits to actual input bytes, require honest alternate-source provenance, preserve stable identities and isolate experiment outputs. One validated loader checks run IDs, data/model hashes, ordered schemas, labels and score contracts. Completed runs preserve every fold, warning and failed trial.

Data and audio limits

Artist strings do not fully resolve aliases or guest performers. Three sampled positive labels have secondary corroboration, while primary chart/recording verification remains incomplete. No labels changed. Waveform-feature parity remains unverified. The new audio and embedding interfaces have software tests, but no real-song embedding experiment has run.

Reproducibility

Run identity: v2_nested_001. Configuration, folds, trials, predictions, uncertainty, package versions and model hashes are preserved in its immutable directory. The baseline archive and exact executed CSV-run source snapshot are retained. Implementation and materials use Codex assistance.

Measured results and interpretation

Run **v2_nested_001**. Final development candidate: **lr_mi_50**. lr_mi_50 records 52.9% balanced accuracy for nested development. Full-development mean tuning balanced accuracy: 52.9%. Nested predictions assess the selection procedure, rather than the single final refit.

Metric	Nested Development	Strict >75%
Accuracy	52.9%	Fail
Balanced Accuracy	52.9%	Fail
Precision	51.6%	Fail
Recall	50.0%	Fail
F1	50.8%	Fail

The approximate 95% group-bootstrap interval (48.3%-57.3%) includes 50%; a reliable advantage over chance is not established.

The five-metric target is not met in this evaluation. Fresh-test confirmation is unavailable.

Matched historical comparison

Historical benchmark	V1 original	V2 candidate
Accuracy	46.8%	55.2%
Balanced Accuracy	46.6%	55.2%
Precision	45.2%	54.8%
Recall	36.8%	52.6%
F1	40.6%	53.7%

The frozen candidate scored 55.2% balanced accuracy on the same 154 historical songs from 19 groups. The paired change from V1 is +8.5 percentage points, with a group-bootstrap interval of -1.6 to +16.9 points. This interval includes zero. These already known records provide a historical comparison, not fresh confirmation.

Demonstration and next inputs

The demo keeps the original model active and labels the V2 option as a research candidate. It shows run identity, evaluation status and strict target checks. Uploaded audio stays exploratory. Next work needs permitted recordings, verified chart/recording identities and a genuinely fresh, locked evaluation collection. A new feature or population study requires its own baseline.

References

- [1] Eric Liu, CS229, 2021: [reference report](#).
- [2] Antonios Malak [feature dataset](#), pinned commit 838e76f, Apache-2.0.
- [3] [Scikit-learn nested cross-validation](#). Local run manifests and predictions supply all reported metrics.